

SATQUERY AI — REMOTE SENSING MISSION REPORT

Indian Space Research Organisation (ISRO) | SIH 26167 Intelligence Dispatch

Natural Language Query:	Locate all oil storage tanks in this image and return coordinates.
Identified Workflow Intent:	GROUNDING
Dispatched Specialist:	GroundingEngine
Composite Confidence:	94.0% (HIGH)

1. Executive Analysis & Findings

Assessment: Visual Grounding Specialist localized 1 spatial feature instance(s) matching 'Locate all oil storage tanks in this image and return coordinates.'. Coordinates and bounding masks have been plotted on the GIS canvas.

2. 4-Signal Multi-Criteria Confidence Matrix

Metric Signal	Component	Score (%)	Status
C_model	Specialist Model Inference Certainty	93.0%	Optimal
C_sensor	Sensor Radiometric SNR & Quality	90.0%	Optimal
C_alignment	Spatial Co-Registration (CRS & IoU)	98.0%	Verified
C_resolution	Ground Sample Distance Suitability	95.0%	Optimal

3. Auditable Agent Execution Pipeline Trace

Step	Action / Milestone	Tool Executed	Latency	Status
1	Raster Ingestion & Metadata Parsing	GeoTIFFProcessor	0.8 ms	COMPLETED
2	Agentic Query Classification & Tool Rout	AgentClassifier	0.0 ms	COMPLETED
3	Specialist Model Execution (GroundingEng	GroundingEngine	0.0 ms	COMPLETED
4	Evidence Fusion & 4-Signal Confidence Ra	CompositeConfidenceEngine	0.0 ms	COMPLETED